Jumana P A

Software Engineer — Backend Development & Agentic AI Engineering

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Professional Summary

Software Engineer with hands-on experience in backend development, AI-powered applications, and agentic workflow development. Experienced in Python, FastAPI, Django, REST APIs, and integrating Large Language Models (LLMs) using the Microsoft Agentic Framework. Worked on developing multi-agent workflows, document intelligence solutions, and automated data extraction pipelines. Familiar with PostgreSQL, Elasticsearch, asynchronous processing, and containerized application development. Interested in building scalable software products and intelligent AI-driven solutions.

Professional Experience

Software Engineer **Jan 2025 – Sep 2026**

Imaginist.ai Techno Solutions Pvt Ltd | AI Product Startup

- Developed and maintained backend services, application features, and REST APIs using Python, FastAPI, and Django.
- Contributed to the development of agentic AI applications and multi-agent workflows using the Microsoft Agentic Framework and Large Language Models.
- Integrated LLM-based capabilities into application workflows to automate business processes and intelligent task execution.
- Developed document intelligence and data extraction workflows to process unstructured documents and transform extracted information into structured data.
- Integrated Google Cloud Document AI and OCR-based processing components for document parsing, field extraction, and data processing.
- Implemented data validation, structured field mapping, error handling, and information indexing for automated document-processing workflows.
- Worked with PostgreSQL and Elasticsearch for application data management, search, and information retrieval.
- Integrated Redis, Celery, and WebSockets to support asynchronous task processing, background execution, and application communication.
- Implemented PII detection and anonymization workflows using Microsoft Presidio to protect sensitive information in AI-powered applications.
- Contributed to frontend development and API integration using HTML, CSS, JavaScript, and React.
- Participated in debugging, code reviews, application optimization, and CI/CD activities while collaborating across different application components.

Internship Experience

Software Development Intern **Oct 2024 – Dec 2024**

Apto Innovations

- Contributed to the development of an Event Management Portal using Laravel and MySQL.
- Implemented application functionality and database-backed features to support event management workflows.
- Worked on backend logic, database operations, debugging, and application components.

Technical Skills

Programming Languages: Python, JavaScript, SQL, Java, C, PHP

Backend Development: FastAPI, Django, Flask, Laravel, REST APIs, API Integration, WebSockets, Asynchronous Programming, Celery

AI & Agentic Development: Microsoft Agentic Framework, LLM Integration,Autogen Framework, AI Agents, Agentic Workflows, Prompt Engineering, Retrieval-Augmented Generation (RAG)

AI & Document Intelligence: Google Cloud Document AI, OCR, Document Processing, Data Extraction, Microsoft Presidio, PII Anonymization

Frontend Development: HTML, CSS, JavaScript, React

Databases & Search: PostgreSQL, MySQL, Redis, Elasticsearch

DevOps & Tools: Docker, Git, GitHub, CI/CD, Linux

Cloud & Infrastructure: Basic familiarity with AWS, Google Cloud Platform (GCP), and Kubernetes

Software Engineering: Object-Oriented Programming, Data Structures and Algorithms, Debugging, Error Handling, Code Reviews, Problem Solving

Academic Project

CodeClarify – AI-Powered Code Understanding Platform

- Developed an academic platform for source-code analysis, summarization, and explanation using Microsoft CodeBERT and Natural Language Processing.
- Fine-tuned the pretrained CodeBERT model using code and natural language datasets to generate automated code explanations.
- Developed backend functionality to process code submissions and generate code explanations.
- Built a Docker-based compiler service to compile and execute Python, C++, and Java code with automated test-case evaluation.
- Worked on dataset preprocessing, model fine-tuning, backend integration, and application development.

Education

Master of Computer Applications (MCA) **2022 – 2024**
APJ Abdul Kalam Technological University, Kerala
CGPA: 8.44

Bachelor of Computer Applications (BCA) **2019 – 2022**
University of Calicut, Kerala
CGPA: 8.00

Certifications

- Programming, Data Structures Using Python – NPTEL
- Cloud Computing – NPTEL